



The thermal neutron monitor with the $^{10}\text{B}_4\text{C}$ converter for the spin-echo SEM spectrometer

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ABSTRACT

A prototype of thermal neutron beam monitor based on a multiwire proportional chamber design with a thin-film converter B_4C has been developed and constructed. The prototype was tested on the ^{252}Cf neutron source, characteristic amplitude spectra were obtained. The possibility of a reliable rejection of unwanted low-amplitude events was demonstrated using the ^{90}Sr β -source.

Based on the prototype, two identical neutron beam monitors with the required parameters for the SEM neutron resonance spin-echo spectrometer were manufactured. This instrument is now under construction and will be part of the instrument suite of the high-flux reactor PIK in Gatchina, Russia.

1. Introduction

A neutron beam monitor is a detector for measuring the intensity of the direct beam in neutron scattering instruments. Monitors are used to normalize physical data relative to the time-varying power of the source, to adjust the neutron guide system and other parts of instruments.

Monitors must meet the following requirements: 1) minimum perturbation of the beam (absorption or scattering), 2) the detection efficiency (and its stability) is defined for specific neutron wavelength and beam intensity, 3) extremely low sensitivity to the γ -background of the source and to the surrounding background to keep the number of measured counts proportional to the neutron flux, 4) a dynamic range that corresponds to the range of beam intensities.

Multiwire proportional chambers (MWPC) are a frequently used technology for neutron monitors due to their reliability and versatility. MWPC has a small amount of material in the beam and can provide low γ -sensitivity.

For gaseous converter, it's very important to ensure the stability of its partial pressure in the detector volume, since changes in the pressure, e. g. due to gas leakage, lead to the drift in detection efficiency.

^3He -filled monitors require both a very high quality of gas sealing and a very small amount of converter gas due to a high cross section of the $^3\text{He}(n,p)\text{T}$ reaction of ~ 5400 b ($1.8 \text{ \AA}$). Gas mixtures are usually

prepared by multiple dilutions with ^4He to adapt the gas mixture to a high flux.

Gas mixtures with a nitrogen converter are often used in monitors designed for high beam intensities ($>10^6$ n/cm²/s), since the $^{14}\text{N}(n,p)^{14}\text{C}$ reaction has a low cross-section of 1.9 b ($1.8 \text{ \AA}$) and an efficiency of 10^{-5} can easily be achieved. The main part of the output energy equal to 577 keV is carried by the proton, while only 43 keV is carried by ^{14}C . Therefore, some of the events, when neutrons interact near cathodes and the proton energy is lost completely or partially (the so-called "wall effect"), and carbon moves from the cathode into the working volume of the detector, are characterized by a noticeable part of low amplitudes in the energy spectrum. It has been noted that the efficiency of amplitude rejection of the γ -background may be worse than when using a ^3He converter [1].

ORDELA [2] and Mirrotron [3] are well-known manufacturers of neutron beam monitors with $^{14}\text{N}_2$ and ^3He converters.

Ionization (fission) chambers with a layer of ^{235}U -based converter (reaction $^{235}\text{U}(n,f)3n$ with the cross section of 681 b at $1.8 \text{ \AA}$) are rather commonly used due to their simple construction the lowest sensitivity to the γ -background: amplitudes of events produced by fission fragments have much larger values.

However, as indicated in Ref. [4], the scattering and absorption reactions in the materials of the monitor made by LND [5] contribute

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significantly ($\geq 7\%$) to a reduction of the neutron beam intensity and result in extra neutron background. Furthermore, the fast neutrons are emitted when uranium undergoes fission, so additional shielding must be provided, which is not always possible to implement in practice. There are some features of fission chamber mentioned in Ref. [6]: nonlinear neutron transmission over the monitor area and nonlinear efficiency behavior that most likely come from an imperfect deposition of the uranium converter. Fission monitors are devices that contain radioactive material and hence require specific storage and radiation control.

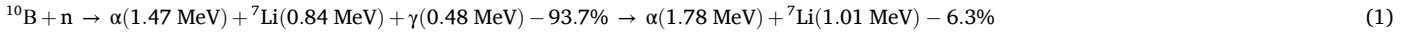
In light of the “ ^3He crisis” [7,8], the last 15–20 years have seen a rapid development of detectors based on the ^{10}B solid converter. The development of new detectors is driven not only by the ^3He supply problem, but also by increasing requirements for detector characteristics for new high-intensity neutron research sources. Search for alternative techniques and many years of research efforts have resulted in a range of promising detectors with neutron converters based on boron carbide B_4C [9–14]. At present, the development of a variety of such detectors is well underway not only for neutron science instruments, but also for radia-

conventional FR4 plate with a soldered grid of gold-plated tungsten wires with a diameter of $30\ \mu\text{m}$, the wire spacing is $s = 2\ \text{mm}$. All wires are joined in the single readout channel. The “anode-cathode” spacing is $L = 2.5\ \text{mm}$. Anode plane is centered between two cathodes, which are the entrance windows of the detector.

The aluminum plate (with thickness of $0.4\ \text{mm}$) with a $\sim 0.3\ \mu\text{m}$ thick converter layer is glued with epoxy resin to the aluminum frame. The B_4C is enriched with ^{10}B to $\sim 90\text{--}95\%$. This assembly forms the cover of the monitor gas volume and the first cathode. The second cathode is located on the base of the chamber and is a single piece fabricated on a milling machine. The thickness of the window is $0.5\ \text{mm}$. Thus, the total thickness of the windows in the prototype is $0.9\ \text{mm}$. The detector has two gas ports for a continuous flow of the gas mixture.

2.2. The neutron detection principle

The conversion reaction of neutron with ^{10}B nuclei occurs in two channels:



tion monitoring systems [15].

Neutron detectors with boron carbide converter outperform ^3He detectors in counting rate capability and design flexibility, and can be used in constructing leading-edge research instruments [16,17]. Due to the great potential of neutron detectors with the B_4C converter and new possibilities for applications on intense beams, we are conducting research work to implement this technology at the PIK high-flux reactor facilities. In this paper, the first results on the development and testing of a monitor for the spin-echo spectrometer SEM [18] based on converters manufactured at the Dubna State University are presented.

2. Details of construction and main characteristics of the monitor

2.1. Monitor construction

The prototype monitor is based on an MWPC with the $60 \times 30\ \text{mm}$ entrance window, see Fig. 1. The anode is a frame made of a

The products of the reaction (1) must leave the B_4C converter layer, enter the working volume of the MWPC to be detected.

Table 1 show the ranges of products from reaction (1) in B_4C . It can be seen that the thickness of the converter should not exceed $3\ \mu\text{m}$ for particles to be able to leave the converter layer. However, as will be shown below, in transmission-type detectors (monitors), the converter should be significantly thinner, so the escape probability of a particle from the converter will be close to 100% .

In laboratory tests of the prototype monitor, a “thick” converter with a B_4C thickness of $\sim 0.3\ \mu\text{m}$ was used*, so that the acquisition time

Table 1

Ranges of products of reaction (1) in B_4C , density $2.52\ \text{g/cm}^3$, calculated in SRIM2008 [19].

Particle energy, MeV	$\alpha(1.47)$	$^7\text{Li}(0.84)$	$\alpha(1.78)$	$^7\text{Li}(1.01)$
Range $\pm$ rms, μm	3.34 ± 0.07	1.68 ± 0.08	4.13 ± 0.07	1.89 ± 0.09

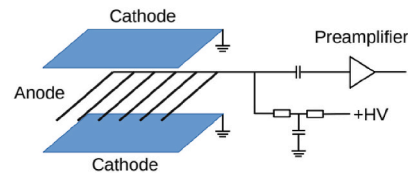
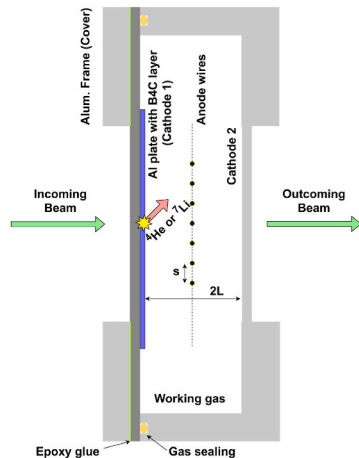


Fig. 1. Schematic of the detector:

(left) MWPC with the B_4C converter located on the entrance window.

(right) Schematic of the signal pickup from the anode and the high-voltage power supply.

would be reasonable with a weak ^{252}Cf neutron source. Therefore, the energy absorption of the α -particle and ^7Li in the converter leads to a partial overlap of the particle energy peaks on the amplitude spectra, Fig. 5.

A converter with a significantly thinner layer of $^{\text{nat}}\text{B}_4\text{C} < 50$ nm is planned for future beam tests, see par. 2.3. Thus, the energy absorption in the converter will be negligible.

* Boron carbide was deposited on the substrates by magnetron sputtering. The thickness of the converter layer was controlled using silicon wafers. The layer thickness was measured on a transverse fracture of the wafer using an electron microscope.

2.3. Conversion probability and detection efficiency

Let us assume that the concentration of molecules in B_4C is:

$$n = \rho \times N_a / M \quad (2)$$

where $\rho = 2.52 \text{ g/cm}^3$ is the density of B_4C ; $N_a = 6.02 \times 10^{-23} \text{ mol}^{-1}$ the Avogadro number; $M = 55.26 \text{ g/mol}$ the molar mass of B_4C . Then for $^{\text{nat}}\text{B}_4\text{C}$ with naturally occurring boron comprised of 19 % ^{10}B , the concentration of active nuclei is $n(^{10}\text{B}) = 2.1 \times 10^{22} \text{ cm}^{-3}$.

Then the neutron conversion efficiency taking into account losses in the converter will be as follows:

$$\varepsilon = \eta \times (1 - \exp[-n \times \sigma(\lambda) \times x]), \quad (3)$$

where $\sigma(\lambda) = \sigma_0 \times \lambda / \lambda_0$ is the cross section of the reaction $n(^{10}\text{B}, \alpha)^7\text{Li}$ for neutrons with the wavelength λ ($\lambda_0 = 1.8 \text{ \AA}$, $\sigma_0(\lambda = \lambda_0) = 3840 \text{ b}$), x is the converter thickness, η is the probability of reaction products leaving the converter layer. The result of calculating the efficiency is shown in Fig. 2.

2.4. Estimation of the dynamic range of measured beam intensities

The maximum neutron beam flux - F_{max} ($\text{n/cm}^2/\text{s}$) that can be detected by the monitor is limited by local counting capability* of an MWPC - R_{max} ($\text{event/cm}^2/\text{s}$ or Hz/cm^2) and is given by

$$F_{\text{max}} = R_{\text{max}} \times \varepsilon(\lambda, d), \quad (4)$$

where $\varepsilon(\lambda, d)$ is the efficiency of the converter with thickness d for a given neutron wavelength λ .

* This refers to a set of parameters: chamber geometry, charge carrier mobility, gas composition, voltage, etc.

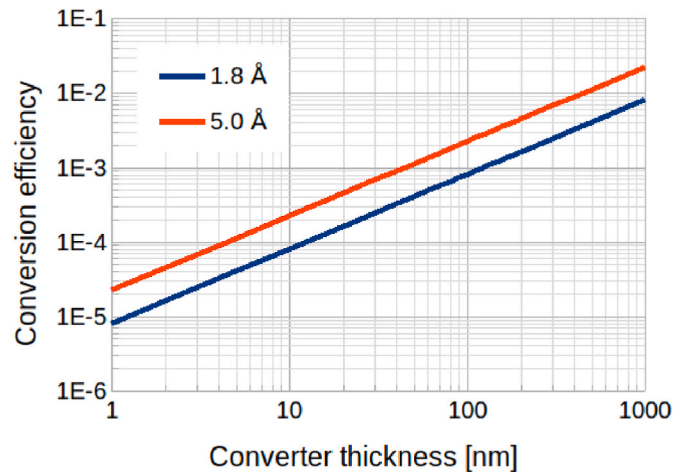


Fig. 2. Calculated neutron efficiency for neutrons with $\lambda = 1.8$ and $5 \text{ \AA}$ with converter $^{\text{nat}}\text{B}_4\text{C}$ (19 % of ^{10}B), $\eta = 1$. The wavelength $\lambda = 5 \text{ \AA}$ corresponds to the maximum in the calculated neutron spectrum at the output channel of the SEM spectrometer.

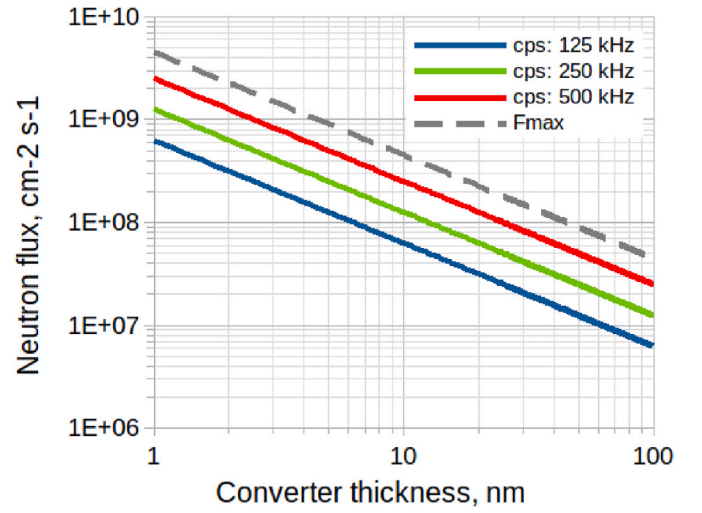


Fig. 3. The estimation of the maximum detectable neutron flux (a neutron wavelength of $5 \text{ \AA}$) at different counting rate capabilities (cps) of the readout electronics, taking into account that the anode wires are combined into one counting channel.

The value of R_{max} is limited by the formation of a space positive shielding charge on the anode under high fluxes. It leads to a local gas gain reduction and, as a result, signal amplitude decrease [20,21].

In order to estimate F_{max} , let us assume $R_{\text{max}} = 1 \text{ kHz/mm}^2$ as counting capacity of similar MWPCs [22,23], then with the aid of formula (3-4), we can find the converter thickness ($^{\text{nat}}\text{B}_4\text{C}$), with which the detector will operate without the gain degradation, see Fig. 3. Here one can also find the estimates of the maximum detectable neutron fluxes at different counting capabilities of the readout electronics.

So, for a monitor to operate with a neutron flux of $\sim 10^8 \text{ n/cm}^2/\text{s}$ and beam cross-section of $S = 3 \times 3 \text{ cm}^2$, the $^{\text{nat}}\text{B}_4\text{C}$ converter thickness should be $\sim 10 \text{ nm}$, and the counting rate capability of the readout electronics should be not less than 200 kHz.

The lower limit of the dynamic range is determined by the background conditions, i.e. the influence of neighboring beams, the reactor background, and the sensitivity of the monitor to the reactor γ -background. The combination of these factors determines the lower intensity limit under real reactor background conditions.

2.5. The choice of the gas mixture

The gas mixture was chosen mainly for four reasons: 1) high drift velocity of charge carriers; 2) low working voltage; 3) possibility of continuous gas-mixture flow without the need for disposal of exhaust gas; 4) it represents a cost-effective and stable solution.

Argon mixtures with a small CO_2 content meet all the above requirements [24]. In Refs. [25–27], detectors with the B_4C converter operate with a different ratio and pressure of the Ar/CO_2 gas mixture, and pure CF_4 is also used. However, Ar/CO_2 mixtures allow operation at lower voltages than CF_4 and achieve γ -sensitivity levels of $\sim 10^{-7}$ or less [28–30]. Therefore, a mixture 90 %Ar + 10 % CO_2 was chosen for tests.

However, the operation of the monitor with pure CF_4 , which has a high stopping power, was also studied, since it is of interest for future tasks of using this gas in profilometers, i.e. monitors with position sensitivity [31]. The position resolution of $\sim 4 \text{ mm}$ can be obtained at $P(\text{CF}_4) = 1 \text{ bar ab}$. It is also worth noting that due to the high Global Warming Potential (GWP) of CF_4 , if there is a need for long-term use of the detector, it is advisable to collect and dispose of the gas without direct release into the atmosphere.

2.6. Secondary particle properties, primary charge estimation, frontend requirements

The number of electron-ion pairs n_{pairs} produced by a particle in the gas is:

$$n_{\text{pairs}} = \Delta E / W,$$

where ΔE is the absorbed energy, $W = [f(\text{Ar})/W(\text{Ar}) + f(\text{CO}_2)/W(\text{CO}_2)]^{-1}$ is the average energy to produce one electron-ion pair, $W(\text{Ar})$ and $W(\text{CO}_2)$ the corresponding values for pure Ar and CO_2 , respectively, and their fractions $f(\text{Ar})$ and $f(\text{CO}_2)$.

So, for the gas mixture 90 % Ar + 10 % CO_2 : $W = [0.9/26 + 0.1/33]^{-1} = 26.56$ eV, where $W(\text{Ar}) = 26$ eV and $W(\text{CO}_2) = 33$ eV from Ref. [32].

The energy lost by the α -particle or ${}^7\text{Li}$ in the gas gap of the MWPC determines the primary ionization charge. Table 2 shows the estimates of the primary ionization charge, produced by products of the reaction ${}^{10}\text{B}(n, \alpha){}^7\text{Li}$. The reaction products produce sufficiently strong primary ionization in both gas mixtures, so the anode charge of ~ 0.1 pC is achieved at a gas gain $G \sim 20$. When using the argon mixture in a 5-mm thick MWPC, there is a partial loss of charge. In the case of CF_4 , the tracks are fit completely into the gap and the entire primary ionization charge is collected.

The calculated gas gain G for the test gas mixtures are shown in Fig. 4. It is seen that for the argon mixture the value $G = 20$ (at which the charge at the anode of 0.1 pC is reached) corresponds to the voltage $HV_A = 1000$ V. The corresponding operating point for CF_4 is significantly higher and is $HV_A \approx 1700$ V.

3. Laboratory tests of the prototype

Tests with the 90 %Ar + 10 % CO_2 mixture and pure CF_4 demonstrate similar results:

- 1) Both gas mixtures exhibit typical amplitude spectra with prominent peaks of α -particle and ${}^7\text{Li}$ absorption [28,34–36], see Fig. 5.
- 2) Low-energy events that correspond to the absorption of β -particles from the ${}^{90}\text{Sr}$ source and γ -particles from the ${}^{252}\text{Cf}$ source are within the low-amplitude range and can be rejected by the discriminator threshold, see Figs. 6–7.

But in the case of pure CF_4 , the operating voltage range is higher by about 700 V, see Fig. 6.

Leaving the 0.3 μm thick converter, the α -particle and ${}^7\text{Li}$ lose some part of their initial energy (see Fig. 8). Moreover, not all α -particle ranges are within the gas gap of 5 mm, see Table 2. This causes the

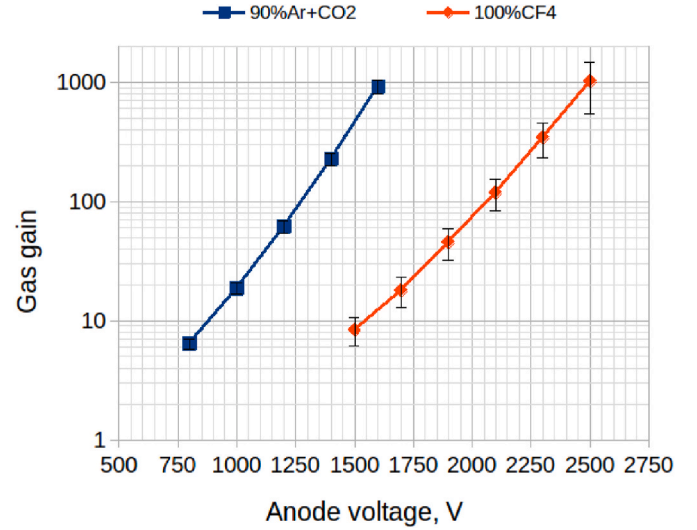


Fig. 4. The gas gain for 90 %Ar+10 % CO_2 and pure CF_4 calculated by Garfield software [33].

measured amplitude spectrum to be continuous from zero to maximum energy.

A detailed analysis and description of the secondary particle spectra is given in Ref. [35]. The structure of spectra is as follows (see Fig. 5): the peak of the total energy of the α -particle 1.47 MeV lies in the region 3000–4000 channels of analog-to-digital converter (ADC), the “tail” 5000–6000 channels of ADC corresponds to rare events of total absorption of the α -particle of the non-primary channel of reaction (1) with energy 1.78 MeV, two peaks in the region 500–2000 channels of ADC belong to ${}^7\text{Li}$ with the energy 0.84 MeV.

The spectra were measured with a spectroscopy charge-sensitive preamplifier Canberra 2006E (charge sensitivity is 47 mV per 10^6 e-ion pairs) and a spectroscopy shaping amplifier ORTEC 517 (Gain = 1k, shaping time 1 μs). The electronic threshold is 50 mV (ADC) and constant. A spectrometry amplitude-to-digital converter ADC-USB-8K-V1 was used [37].

Neutrons from a ${}^{252}\text{Cf}$ source can be registered already at the voltages of 50–100 V, see Fig. 6. This means that the MWPC operates as ionization chamber without amplification. As can be seen in Fig. 5, the amplitude spectrum is continuous, so the count increases smoothly up to 800 V, with the bulk of the ionization charges from α -particles and ${}^7\text{Li}$ detected.

It is important to note that the ${}^{252}\text{Cf}$ source produces a significant flux of γ -rays, which can subsequently be absorbed in the detector,

Table 2

Characteristics of secondary particles for gases 90 %Ar + 10 % CO_2 and pure CF_4 . Ranges of particles and ionization losses were calculated by SRIM2008 software package. The average energy of production of electron-ion pair for CF_4 is $W = 33.8$ eV.

Gas mixture:		90 % Ar + 10 % CO_2					100 % CF_4		
Channel of reaction (1)	E_0 , keV	R, mm	ΔE^a , keV	$n_{\text{pairs}} \times 10^3$	q_0 , fC		R, mm	$n_{\text{pairs}} \times 10^3$	q_0 , fC
1: 93.7 %	α	1470	8.0	1043.8	39.3	6.3	3.2	43.2	6.9
	Li-7	840	4.2	819.7	30.9	4.9	1.8	24.0	3.8
	γ	480							
2: 6.3 %	α	1780	9.8	948.2	35.7	5.7	3.8	52.2	8.4
	Li-7	1010	4.7	986.6	37.1	5.9	2.0	29.2	4.7

The following notations are used.

E_0 - the initial energy of the particle.

ΔE - the ionization energy absorbed in the MWPC gap (5 mm).

R - the full particle range in Ar/ CO_2 gas mixture.

n_{pairs} - the number of e^- /ion pairs produced in the MWPC gap.

$q_0 = n_{\text{pairs}} \times e^-$, $e^- = 1.6 \times 10^{-19}$ C - the primary charge.

^a The calculations considered the simplified case when α -particles escape the converter perpendicular to its plane. In reality, the particles are emitted isotropically from different depths of the converter, so the range of particle energies at escaping will be continuous: from zero to the total energy of 1.47 MeV for the main channel (and 1.78 MeV for the second channel) when the particle escapes at a large angle from the converter surface into the MWPC gap.

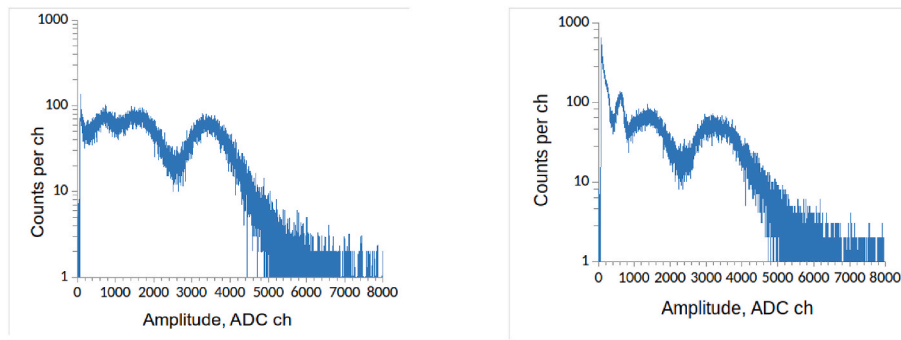


Fig. 5. Amplitude spectra obtained by the monitor prototype (logarithmic scale of the Y-axis):

(left) 90 %Ar+10 %CO₂, HV_a = 800 V, exposition time ~20.4 h,

(right) 100 % CF₄, HV_a = 1650 V, exposition time ~20.5 h.

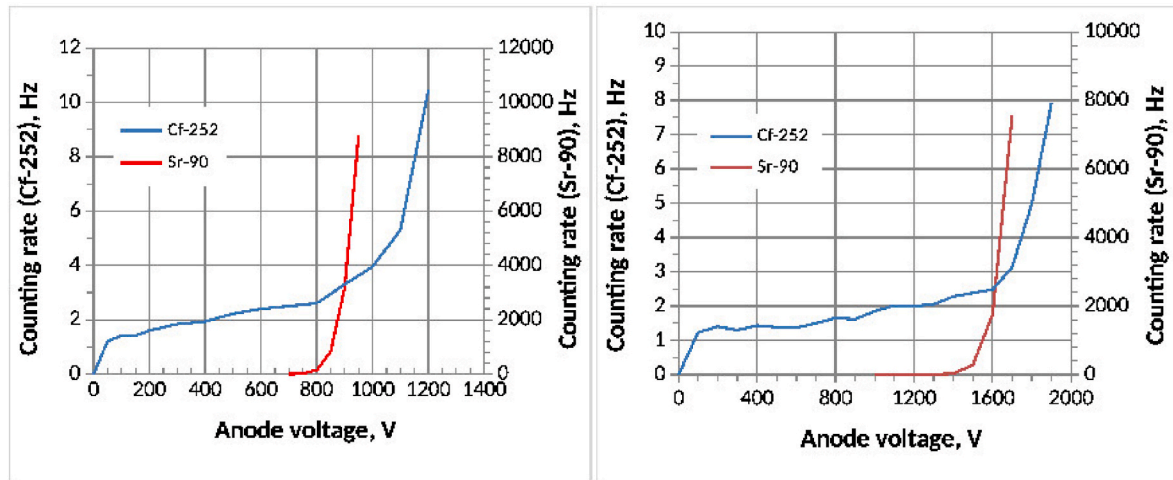


Fig. 6. Counting rates of the prototype operating with 90 %Ar+10 %CO₂ (left) and 100 %CF₄ (right), measured for different sources: ²⁵²Cf (blue line) and ⁹⁰Sr (red line). The radioactivity of ⁹⁰Sr ~75 × 10⁷ Bq or 2 × 10⁷ e⁻/sec. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

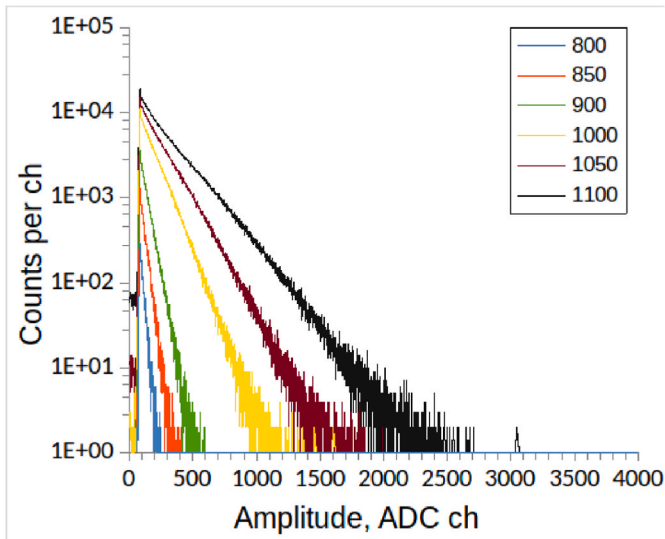


Fig. 7. Amplitude spectra (logarithmic scale of the Y-axis) for different anode voltages were measured by the prototype at ⁹⁰Sr irradiation (radioactivity of ~75 × 10⁷ Bq). The source was placed at a distance of 2 cm from the entrance window. Gas mixture 90 %Ar+10 %CO₂, amplifier (ORTEC 517): gain - 1k, shaping time - 1 μs, exposition time - 60 s, without normalization.

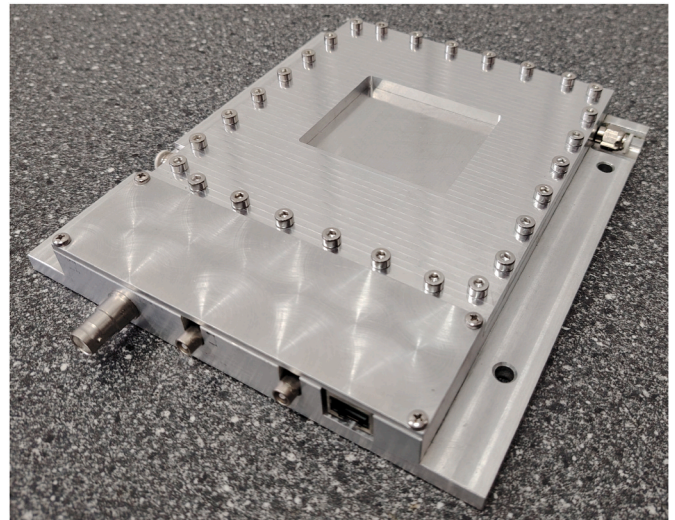


Fig. 8. A photo of the monitor for SEM. The signal connectors, the high-voltage connector and gas ports are visible.

producing low-energy events. As the voltage increases above 1000 V, the monitor also begins to detect low-energy events.

The sensitivity of the prototype to background radiation was tested

by a ^{90}Sr β -source. The amplitude spectrum of electrons has a fundamental difference from the neutron-induced spectrum due to the small primary charge left by fast electrons in the gas. The characteristic spectrum of fast electrons occupies the region of low amplitudes, and so it allows us to detect a weak neutron flux from ^{252}Cf in the background of a strong ^{90}Sr source by rejection of low-energy events.

Counting rates and spectra for ^{90}Sr are given in Figs. 6–7. For the gas mixture 90 %Ar + 10 %CO₂, the events from ^{90}Sr start to be detected from a voltage of ~ 750 V at a minimal threshold of the electronics. When the voltage is further increased, the count increases very rapidly, due to the fact that the spectrum of the $^{90}\text{Sr}/^{90}\text{Y}$ β -source is continuous with the maximum electron energy $E_{\text{max}} = 2.28$ MeV. In the region of working voltages (up to 850 V), β -particles are easily suppressed by the discriminator threshold (for $HV_a = 800$ and 850 V, the thresholds are 250 and 350 mV, respectively), while the neutron counting rate from the ^{252}Cf source remains the same.

For the CF₄ gas, the ^{90}Sr electrons start to be detected at the voltage of ~ 1300 V.

4. Monitor for the spin-echo SEM spectrometer

The calculated spectrum of thermal neutrons at the SEM instrument has a mean wavelength after the velocity selector of $5 \text{ \AA}$ with the expected neutron rate is 10^9 n/s. To adapt the MWPC and electronics to such intensities, the conversion efficiency must be $\sim 10^{-4}$, which can in turn be achieved when the thickness of $^{nat}\text{B}_4\text{C}$ is $\sim 5\text{--}10$ nm, see Fig. 2. In this case, the loading of the readout electronics is $\sim 200\text{--}300$ kHz. With this in mind, the charge-sensitive preamplifier-discriminator based on the n-JFET transistor BF862 was developed for the monitor. The characteristics of the preamplifier are given in Table 3.

Monitor was tested using the 90 %Ar + 10 %CO₂ gas mixture. The B₄C converter of $\sim 1 \mu\text{m}$ thick was used in order to reduce the statistics collection time. The counting rate has increased three times compared to the prototype.

Fig. 9 shows the amplitude spectra obtained at the ^{252}Cf neutron source for voltages of 750, 800 and 850 V. It can be seen that the use of a thicker converter slightly changes the spectrum, energy peaks are less clearly defined. This is due to the increased energy loss when particles leave the B₄C layer, see also [35]. The contribution of α -particles with the energy 1.77 MeV, which correspond to the region of channels 1100–1300 ($HV_a = 800\text{V}$), becomes noticeable at higher statistics, see Fig. 9.

Fig. 10 illustrates the measured counting characteristics of the monitor. It can be seen that the counting rate is steeper than that of the prototype, which is due to the shape of the amplitude spectrum (Fig. 9). The count of neutrons starts, as in the prototype, at low voltages of ~ 100 V and then smoothly increases up to 900 V. No obvious counting plateau is observed, since the energies of secondary particles lie in the range from zero to maximum. The working voltages are in the range

Table 3

The main characteristics of the frontend electronics for the SEM monitor.

Charge sensitivity	40 mV/fC
Noise (ENC), rms	1500e (0.25 fC) + 20 e/pF
Analog output signal (could be selected)	- Unipolar (quasi-Gaussian), $T_m = 300$ ns - Bipolar, $T_m = 200$ ns for counting loads up to 300 kHz
Dynamic range	65 fC
Logic output	TTL (RS485), 100 ns
LV supply	$\pm 5\text{V}$, 90 mA
Connectors	1 RJ-45: Analog signal, TTL (RS485), LV $\pm 5\text{V}$, discriminator DC-level 2 Analog output (LEMO) 3 Test input (LEMO)
Test input	Charge coupled to preamp input at 0.5 fC/mV; $C_{in} = 1$ pF, $Z_{in} = 50$ Ohm
PCB dimensions, mm	95 $\times$ 35

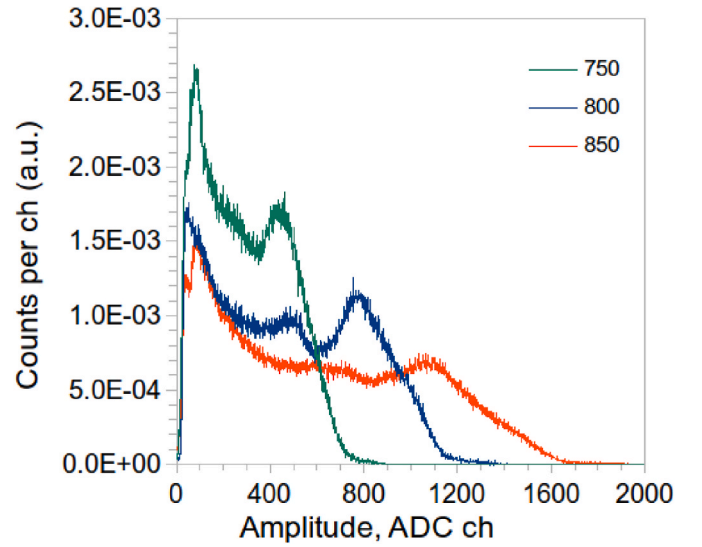


Fig. 9. Amplitude spectra (normalized to the number of events) from the ^{252}Cf source measured by the monitor at various anode voltages HV_a : (green line) 750 V, the counting rate is 7.9 Hz, the exposition time is 17.5 h; (blue line) 800 V, the counting rate is 10.4 Hz, the exposition time is 18.7 h; (red line) 850 V, the counting rate is 10.9 Hz, the exposition time is 25.3 h. The ADC threshold is ~ 5.5 fC (or 30 ch ADC). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

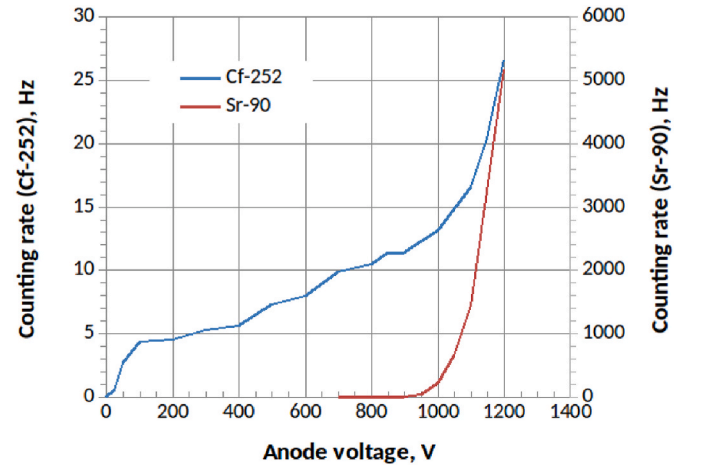


Fig. 10. Counting rates measured with the 90 %Ar + 10 %CO₂ mixture for various sources: (blue line) ^{252}Cf , (red line) ^{90}Sr (radioactivity of $\sim 225 \times 10^7$ Bq or 6×10^7 e⁻/sec). The source was 1 cm away from the monitor window. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

700–850 V.

Results of tests performed at the ^{90}Sr show that at an anode voltage of 850 V the amplitudes of β -particles do not exceed the level of ~ 5 fC. In addition, the detection of neutrons from the ^{252}Cf source is possible when the monitor is simultaneously irradiated with an intense ^{90}Sr β -source (the current from the source was 5 nA), which did not affect the neutron counting rate.

5. Conclusions

The prototype of a thermal neutron beam monitor based on the B₄C converter has been developed and tested on a neutron and β -sources.

The detector has similar amplitude spectra for both 90 %Ar+10 %CO₂ and 100 %CF₄ gas mixtures. The argon-based mixture allows working at lower voltages. The amplitude spectrum of β -particles lies in the low-amplitude region, which can be rejected without a considerable loss in neutron counting rate.

Our results enabled us to construct a full-scale monitor for the spin-echo spectrometer SEM. In order to optimize the converter thickness, parameters of electronics, and to study the monitor operation in real background conditions of the reactor, further work will be aimed at testing the monitors on high-intensity monochromatic thermal neutron beams at IR-8 (NRC “Kurchatov Institute”, Moscow) or IBR-2 (JINR, Dubna). For this purpose, converters based on B₄C with thickness ranging up to 50 nm will be fabricated. Due to the total thickness of the windows of the final monitor of 0.65 mm, the very high neutron transmission is expected.

Dedicated to the memory of Yulia Ilyina.

CRedit authorship contribution statement

D.S. Ilyin: Writing – original draft, Methodology, Investigation. **B.V. Bochin:** Investigation, Conceptualization. **V. Yu Egorov:** Investigation. **V.V. Ivanov:** Methodology. **A.G. Kolesnikov:** Resources, Methodology. **K.A. Pshenichnyi:** Software. **K. Yu Terentiev:** Investigation. **V.V. Tarnavich:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: V. V. Tarnavich reports financial support was provided by Russian Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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